

# Supplementary Information

Dissecting Sequence, Structure, and Data Recency Effects  
in Enzyme Commission Prediction under Temporal Distribution Shift

Seunghyon Kim

School of Applied AI and Entrepreneurship, Handong Global University

Pohang, 37554, Republic of Korea

jamesksh0130@gmail.com

ORCID: 0009-0002-0708-748X

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## S1 Related Work

**Classical sequence alignment.** Before machine learning, enzyme function annotation relied primarily on sequence similarity transfer. BLAST [Altschul et al., 1990] identifies homologous proteins by computing pairwise alignment scores against curated databases such as UniProtKB/Swiss-Prot; a query enzyme is annotated with the EC number of its closest hit above an e-value threshold. On the EC-Bench temporal test set, BLASTp achieves a weighted F1 of 0.502 at the 100% similarity threshold, dropping to 0.431 when training is restricted to proteins with at most 30% sequence identity to the test set [Davoudi et al., 2026]—illustrating the well-known fragility of alignment-based methods for evolutionarily distant queries. Profile-based methods such as PRIAM mitigate this limitation through position-specific scoring matrices derived from multiple sequence alignments, but they cannot integrate non-sequence information such as tertiary structure.

**Deep learning for EC prediction.** DeepEC [Ryu et al., 2019] and DEEPRE introduced 1D CNNs over one-hot amino acid sequences with global average pooling; while effective for EC Levels 1–3, these models struggle at Level 4 due to fine-grained specificity requirements and limited receptive fields. ECPick [Han et al., 2024] augments a hierarchical 1D CNN with Dempster–Shafer evidential theory, assigning a trustworthiness score to each prediction. CLEAN [Yu et al., 2023] reframes EC prediction as metric learning: ESM-1b embeddings are projected into a shared latent space trained with a triplet margin loss. HIT-EC [Dumontet et al., 2026] designs a four-level Transformer achieving micro F1 of 0.93 on Swiss-Prot—the current state of the art for sequence-only EC prediction.

**Protein language model encoders.** ESM-2 [Lin et al., 2023] is a family of masked language models—scaling from 8M to 15B parameters—pre-trained on over 250 million sequences from UniRef90. The 650M-parameter variant produces per-residue representations of dimension 1,280; the [CLS] token embedding encodes a global sequence summary.

**Contact map encoding.** A protein contact map is a binary symmetric matrix  $\mathbf{C} \in \{0, 1\}^{N \times N}$ , where  $C_{ij} = 1$  if the  $C\alpha$  distance between residues  $i$  and  $j$  falls below 8 Å [Gligorijevic et al., 2021]. Contact maps encode tertiary fold topology in a translation- and rotation-invariant form. DeepFRI [Gligorijevic et al., 2021] treats the contact map as a sparse adjacency matrix and propagates per-residue features through graph convolutional layers. CLEAN-Contact [Ayres et al., 2024] uses ResNet-50 to encode 2D contact maps, concatenating the resulting representation with ESM-2 features before contrastive learning; on New-392 and Price-149 this achieves a 12% and 16% F1 improvement over sequence-only CLEAN.

**EC-Bench temporal evaluation.** EC-Bench [Davoudi et al., 2026] addresses temporal data leakage by using all reviewed Swiss-Prot entries as of February 2018 for training and 468 proteins added by January 2023 as the test set. The primary metric is the weighted F1-score at Level 4, reported at similarity thresholds of 30%–100%.

## S2 Extended Dataset Details

**Data splits.** We constructed two test sets: (i) *random split* (80/10/10) using a fixed random seed; (ii) *cluster split* using MMSeqs2 [Steinegger and Söding, 2017] at 30% sequence identity, following EC-Bench [Davoudi et al., 2026] to evaluate OOD generalization.

**Split integrity checks.** Because Swiss-Prot contains closely related and sometimes identical protein sequences under different accessions, we audited all train/validation/test partitions for accession and exact-sequence overlap. The random split contains no accession overlap across train, validation, and test sets, but it does contain exact sequence duplicates across partitions; therefore, we treat random-split performance as in-distribution evaluation. For sequence-disjoint evaluation, we use the cluster split, where cluster-train, cluster-validation, and cluster-test contain zero accession overlap and zero exact-sequence overlap. Cluster-test results are reported only for models trained on the corresponding cluster-train split.

**Multi-label annotation statistics.** EC annotations are naturally multi-label: a single protein may catalyze multiple reactions and therefore receive multiple Level-4 EC numbers. In the main Swiss-Prot training split, 8,385 proteins (3.88%) contain more than one Level-4 EC annotation, with up to nine Level-4 labels for a single protein. The average Level-4 label cardinality is 0.893 per protein when partial EC annotations are included. These statistics motivate sigmoid outputs and binary cross-entropy rather than a single-label softmax classifier.

## S3 Training Dynamics

Figure 1 and Table 1 summarize best-checkpoint training dynamics for the main ablation models. All models were trained for up to 30 Phase-1 epochs with frozen ESM-2 representations; Contact-EC-Hier was then trained for an additional 20 Phase-2 epochs with partial unfreezing. B3 (contact map only) converges in approximately 10 epochs, reaching val-hard L4 micro F1 comparable to B1 (0.7650 vs. 0.7655). This fast convergence suggests that the contact-map branch learns a compact structural signal rather than merely memorizing sequence-derived labels. Contact-EC-Hier improves beyond both single-modality baselines during Phase 1 and continues to gain after partial unfreezing in Phase 2, reaching 0.8698 on val-hard. The hierarchical sequence-only B2 baseline remains substantially lower throughout training, consistent with the temporal and val-hard underperformance reported in the main text.

Table 1: Best epoch and val-hard L4 micro F1 for each model.

| Model                      | L4 Micro F1 (val hard) | L4 Macro F1 (val hard) | Best Epoch   |
|----------------------------|------------------------|------------------------|--------------|
| B1 (ESM-2, flat FC)        | 0.7655                 | 0.1353                 | 30           |
| B2 (ESM-2, hier. FC)       | 0.4204                 | 0.0383                 | 30           |
| B3 (Contact map only)      | 0.7650                 | 0.1335                 | 10           |
| Contact-EC (flat FC, ours) | 0.8879                 | 0.2294                 | 29           |
| Contact-EC-Hier            | 0.8698                 | 0.2141                 | 20 (Phase 2) |

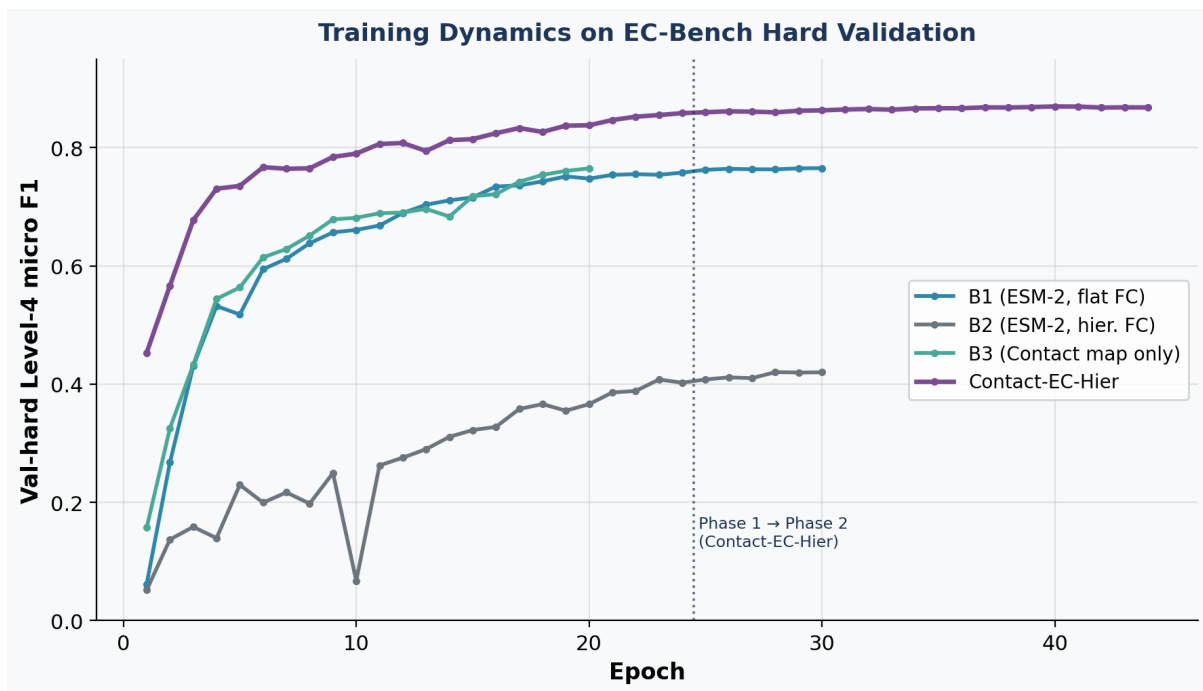


Figure 1: Training dynamics on the EC-Bench hard validation split. Phase 1 uses frozen ESM-2 representations for sequence-containing models; Phase 2 partially unfreezes Contact-EC-Hier. The contact-map-only B3 model rapidly approaches the flat ESM-2 baseline B1, whereas Contact-EC-Hier benefits from sequence–structure fusion and additional Phase 2 optimization.

## S4 Full EC-Bench Leaderboard Comparison

Table S2 reports results on all 468 EC-Bench test proteins, including 309 with partial annotations and 35 with novel EC classes outside the training vocabulary. The all-468 score is reported as an operational closed-set coverage audit, not as a fully defined Level-4 evaluation for partial annotations.

To avoid treating partial EC annotations as Level-4 failures, we additionally evaluate each partial label only up to its deepest annotated hierarchy prefix. For example, EC 3.4.24.- is evaluated through Level 3, whereas EC 1.-.- is evaluated only at Level 1. This prefix-level audit is not directly comparable to complete Level-4 F1, but it quantifies whether the model recovers the known functional hierarchy for incompletely curated proteins.

Contact-EC (weighted F1 = 0.4763) exceeds EnzBert-learned (+1.3 pp) despite training on an earlier corpus with a  $4.5\times$  smaller backbone. Contact-EC-Hier (0.4325) falls 3.8 pp below EnzBert-learned; the gap to Contact-EC (+4.4 pp) reflects the flat FC head’s superior calibration on partial/novel-EC proteins under distribution shift.

The case-wise comparison confirms that HIT-EC is the stronger temporal predictor in absolute terms. Contact-EC uniquely recovers two cases: A0A3G9HRC2, a multi-label oxidoreductase with EC 1.14.14.1 and EC 1.6.2.4, and A0A6C0WW38, where HIT-EC predicts the neighbouring methyltransferase class EC 2.1.1.159 instead of EC 2.1.1.160. Conversely, many HIT-EC-only successes are near misses for Contact-EC: for C0HLB8 (EC 3.1.1.4), Contact-EC assigns EC 3.1.1.4 the highest probability but below the fixed 0.5 threshold (0.4931). These cases support the main text’s interpretation that Contact-EC is useful for decomposing structure and calibration effects, whereas HIT-EC remains the stronger closed-set temporal classifier.

Table 2: Full EC-Bench temporal test: all 468 proteins vs. evaluable 124. The all-468 columns quantify closed-set coverage under label shift; partial-EC proteins are not fully Level-4 scorable, and novel-EC proteins are outside the SP-2018 vocabulary.

| Model                      | All 468       |               | Known-EC 124  |               |            |
|----------------------------|---------------|---------------|---------------|---------------|------------|
|                            | Micro F1      | Weighted F1   | Micro F1      | Weighted F1   |            |
| B1 (ESM-2, flat FC)        | 0.3539        | 0.3156        | 0.4216        | 0.3178        | † EC-Bench |
| B2 (ESM-2, hier. FC)       | 0.1098        | 0.0835        | 0.1111        | 0.0835        |            |
| B3 (Contact map only)      | 0.3167        | 0.2625        | 0.3646        | 0.2734        |            |
| Contact-EC (flat FC, ours) | 0.3716        | <b>0.4763</b> | <b>0.6032</b> | <b>0.5026</b> |            |
| Contact-EC-Hier            | <b>0.3771</b> | 0.4325        | 0.5690        | 0.4528        |            |
| EnzBert-learn†             | –             | 0.463         | –             | –             |            |
| Stacked ensemble†          | –             | 0.524         | –             | –             |            |

leaderboard; weighted F1 on all 468 proteins.

Table 3: Hierarchical prefix evaluation on the 309 partial-EC temporal proteins. Partial annotations are scored only at levels where ground truth prefixes are available. Level-4 is omitted because no complete Level-4 ground truth exists for these rows.

| Model                      | L1 Micro F1   | L2 Micro F1   | L3 Micro F1   |
|----------------------------|---------------|---------------|---------------|
| B1 (ESM-2, flat FC)        | 0.8311        | 0.5805        | 0.4656        |
| B3 (Contact map only)      | 0.7930        | 0.6615        | 0.5933        |
| Contact-EC (flat FC, ours) | 0.8259        | 0.7322        | 0.6629        |
| Contact-EC-Hier            | <b>0.8409</b> | <b>0.7685</b> | <b>0.6975</b> |

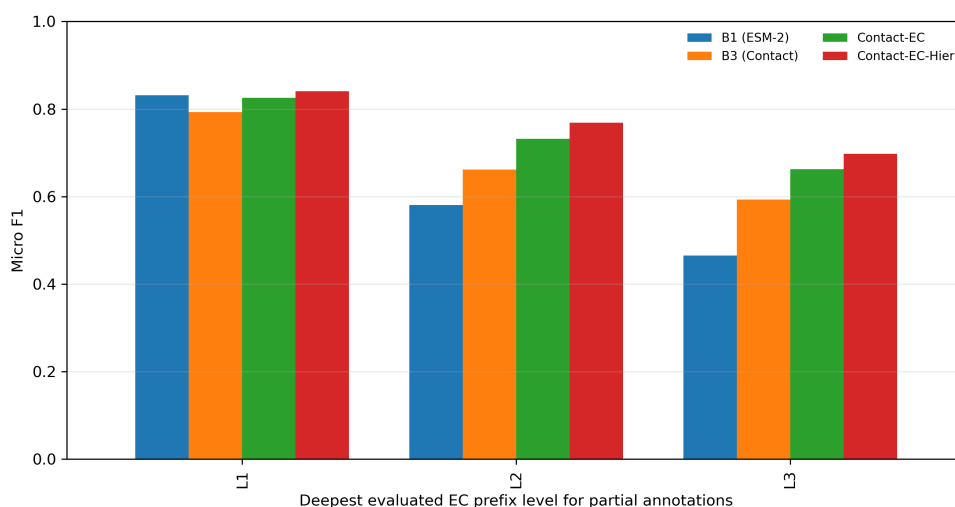


Figure 2: Hierarchical prefix micro F1 on partial-EC temporal proteins. The result shows that partial annotations should be handled as prefix-level supervision rather than assigned Level-4 failures.

Table 4: Case-wise HIT-EC vs. Contact-EC comparison on the 124 complete known-EC Swiss-Prot 2023-01 temporal proteins. A case is counted as correct if the model recovers at least one true Level-4 EC label.

| True L1 | Both correct | HIT-EC only | Contact-EC only | Both wrong | Total |
|---------|--------------|-------------|-----------------|------------|-------|
| 1       | 12           | 5           | 1               | 0          | 18    |
| 2       | 13           | 8           | 1               | 3          | 25    |
| 3       | 29           | 11          | 0               | 2          | 42    |
| 4       | 5            | 20          | 0               | 7          | 32    |
| 5       | 4            | 1           | 0               | 2          | 7     |
| All     | 63           | 45          | 2               | 14         | 124   |

**EC Level-1 family analysis.** To add biological granularity, we recomputed temporal predictions for the ESM-2-only B1 model, the contact-only B3 model, and Contact-EC on the 124 fully evaluable temporal proteins, then stratified Level-4 micro F1 by true EC Level 1 family. Table 5 and Figure 3 show that Contact-EC yields the largest gains over ESM-2 in oxidoreductases (EC 1; +0.6270 micro F1) and hydrolases (EC 3; +0.2753), whereas lyases (EC 4) remain difficult for all Contact-EC variants despite HIT-EC’s stronger performance. These results suggest that the utility of contact-map fusion is enzyme-family dependent rather than uniform across EC space. Because several families contain few temporal examples, the analysis should be interpreted as hypothesis-generating rather than mechanistic evidence about active-site geometry.

Table 5: Level-1 family analysis on the 124 complete known-EC Swiss-Prot 2023-01 temporal proteins. Values are Level-4 micro F1 within each true Level-1 family. EC 6 and EC 7 are absent from this evaluable subset.

| EC | Family          | <i>N</i> | ESM-2  | Contact | Contact-EC | HIT-EC |
|----|-----------------|----------|--------|---------|------------|--------|
| 1  | Oxidoreductases | 18       | 0.0909 | 0.2222  | 0.7179     | 0.9231 |
| 2  | Transferases    | 25       | 0.6667 | 0.6154  | 0.7059     | 0.8649 |
| 3  | Hydrolases      | 42       | 0.5085 | 0.4561  | 0.7838     | 0.9545 |
| 4  | Lyases          | 32       | 0.1923 | 0.0833  | 0.1695     | 0.6667 |
| 5  | Isomerases      | 7        | 0.3636 | 0.2500  | 0.6667     | 0.7143 |

Table 6: Contact-EC threshold sensitivity on the 124 complete known-EC Swiss-Prot 2023-01 temporal proteins. The fixed threshold of 0.5 is used in the main benchmark table.

| Setting                 | Threshold | Micro F1 | Weighted F1 | Precision | Recall | Empty |
|-------------------------|-----------|----------|-------------|-----------|--------|-------|
| Fixed global            | 0.50      | 0.6032   | 0.5026      | 0.7677    | 0.4967 | 45    |
| Best global by micro F1 | 0.65      | 0.6167   | 0.4895      | 0.8506    | 0.4837 | 54    |
| 0.5 or top-1 if empty   | 0.50      | 0.5589   | 0.5413      | 0.5764    | 0.5425 | 0     |
| Top-1 only              | –         | 0.5126   | 0.4628      | 0.5726    | 0.4641 | 0     |

Threshold sensitivity analysis shows that the fixed 0.5 cutoff is not the sole explanation for Contact-EC’s temporal gap. Raising the global threshold to 0.65 slightly improves micro F1 from 0.6032 to 0.6167 by increasing precision, but reduces recall and weighted F1. Lower-threshold or top-1 fallback variants recover more labels but introduce additional false positives. This behaviour is consistent with a

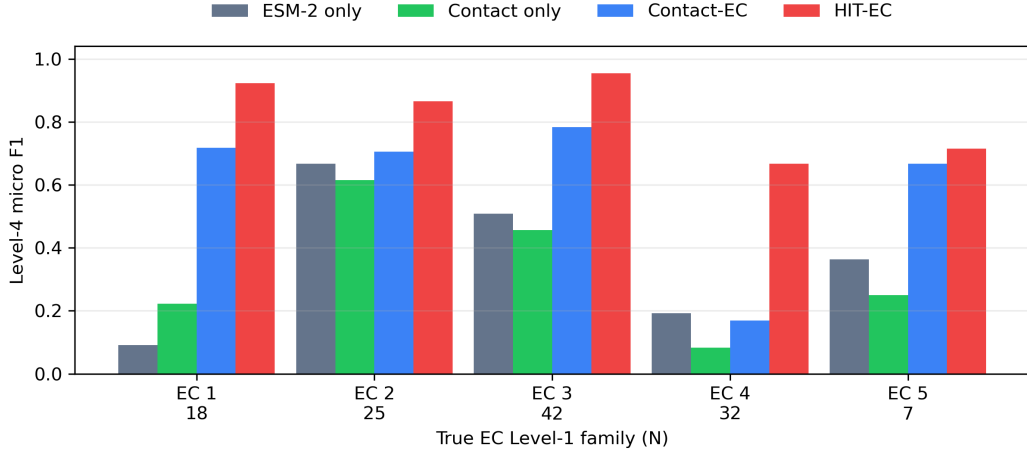


Figure 3: Temporal Level-4 micro F1 stratified by true EC Level-1 family. The largest Contact-EC gains occur in oxidoreductases and hydrolases, while lyases remain a failure mode for the current contact-fusion model.

calibration–recall trade-off rather than a simple thresholding artifact.

## S5 External Benchmark Evaluation

Table 7: External benchmark results (per-protein F1, threshold=0.5). CLEAN / CLEAN-Contact results from Ayres et al. [2024]. ¶: Contact-EC-ExpA on full SP-2026 (243K); New-392 pre-dates SP-2026, creating likely train/test overlap—interpret as upper bound.

| Method                              | Backbone               | Train data | New-392       | Price-149     |
|-------------------------------------|------------------------|------------|---------------|---------------|
| CLEAN-Contact [Ayres et al., 2024]  | ESM-2 3B               | SP-full    | <b>0.566</b>  | <b>0.525</b>  |
| CLEAN [Yu et al., 2023]             | ESM-2 3B               | SP-full    | 0.504         | 0.452         |
| ProteInfer [Sanderson et al., 2023] | CNN                    | UniRef90   | 0.309         | –             |
| B1 (ESM-2)                          | ESM-2 650M             | SP-2018    | 0.170         | 0.026         |
| B3 (Contact)                        | ResNet-50              | SP-2018    | 0.154         | 0.000         |
| Contact-EC (flat FC)                | ESM-2 650M + ResNet-50 | SP-2018    | <b>0.3717</b> | <b>0.1775</b> |
| Contact-EC-Hier                     | ESM-2 650M + ResNet-50 | SP-2018    | 0.358         | 0.096         |
| Contact-EC-ExpA¶                    | ESM-2 650M + ResNet-50 | SP-2026    | 0.7680¶       | <b>0.2161</b> |

**New-392.** Contact-EC (SP-2018) achieves per-protein F1 = 0.3717, above ProteInfer (0.309) but below CLEAN (0.504) and CLEAN-Contact (0.566). The gap has two causes: (i) CLEAN-Contact uses ESM-2 3B ( $4.5\times$  larger) and (ii) contrastive retrieval benefits from training-example similarity at inference. Contact-EC-ExpA (SP-2026) achieves 0.7680 but cannot be treated as a fair comparison since New-392 proteins pre-date 2026 and likely appear in the training corpus.

**Price-149.** Contact-EC (0.2500 micro F1) achieves the highest score among SP-2018 models, but the external Price-149 benchmark remains a major failure mode relative to CLEAN-Contact (0.525). To determine whether this gap is caused by missing inputs, we audited the raw 149 proteins, encoded SP-2018 evaluation subset, label vocabulary, and structure assets. Table 8 shows that input coverage is not the primary explanation: ESM-2 embeddings and nonzero contact maps are available for all 149 raw proteins, and 139 proteins have complete Level-4 EC labels contained in the SP-2018 vocabulary. Instead, the failure is concentrated at exact Level-4 specificity. In the hierarchical diagnostic, Contact-EC retains high coarse EC performance on Price-149 (Level 1 micro F1 = 0.9379; Level 2 = 0.7862; Level 3 = 0.7655), but Level 4 micro F1 falls to 0.0508 (Figure 4). The contact-only B3 model scores 0.0000 on the canonical Price-149 evaluation, and ESMFold-derived contact maps do not recover the gap (Contact-EC ESMFold-map micro F1 = 0.1218). We therefore interpret Price-149 as an external bacterial RefSeq distribution-shift stress test: the model can often identify the broad enzyme family, but its current SP-2018 contact-fusion calibration does not transfer reliably to exact EC subclasses.

Table 8: Price-149 failure audit. Coverage is computed on the raw Price-149 file ( $N = 149$ ); canonical Level-4 model metrics use the encoded SP-2018 evaluation subset ( $N = 136$ ).

| Audit item                                               | Count/value | Interpretation                                  |
|----------------------------------------------------------|-------------|-------------------------------------------------|
| Raw Price-149 proteins                                   | 149         | external bacterial RefSeq benchmark             |
| Encoded SP-2018 subset                                   | 136         | canonical flat Level-4 evaluation subset        |
| Proteins with embeddings                                 | 149         | no missing ESM-2 input bottleneck               |
| Proteins with nonzero contact maps                       | 149         | no missing contact-map bottleneck               |
| Raw proteins with all EC labels in SP-2018 L4 vocabulary | 139         | moderate label-vocabulary coverage              |
| Unique raw Level-4 EC classes                            | 56          | external label diversity                        |
| Unique raw Level-4 EC classes outside SP-2018 vocabulary | 5           | limited closed-set mismatch                     |
| B1 ESM-2 Level-4 micro F1                                | 0.0324      | sequence-only transfer collapses                |
| B3 contact-only Level-4 micro F1                         | 0.0000      | contact-only transfer collapses                 |
| Contact-EC flat Level-4 micro F1                         | 0.2500      | fusion improves but remains below CLEAN-Contact |
| Contact-EC hierarchical Level-1 micro F1                 | 0.9379      | broad enzyme family often recovered             |
| Contact-EC hierarchical Level-4 micro F1                 | 0.0508      | exact subclass specificity fails                |

**Contact-EC calibration and prediction-cardinality audit.** We further audited whether the Price-149 failure is explained by empty predictions at the fixed 0.5 threshold. Table 9 and Figure 5 show that this is not the only explanation. On Price-149, Contact-EC emits nonempty predictions for 51.5% of encoded proteins, with median top-1 probability 0.5527, but the top-1 hit rate is only 27.9% and Level-4 micro F1 is 0.2500. Thus the external failure reflects exact-subclass specificity and calibration under distribution shift, not simply missing structures, absent embeddings, or universally empty prediction sets. For partial and novel temporal proteins, Level-4 F1 and top-1 hit rate are undefined because the closed-set target is incomplete or outside the training vocabulary; for those groups we report only prediction burden and confidence diagnostics. The flattened multilabel calibration errors are therefore used only as diagnostics because they are dominated by the many negative Level-4 labels.



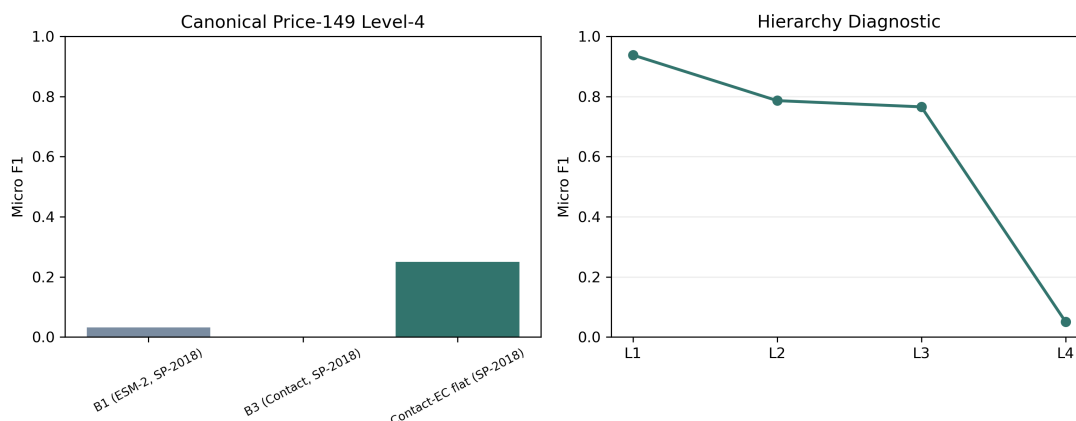


Figure 4: Price-149 failure analysis. Left: canonical Level-4 micro F1 for SP-2018 models. Right: hierarchical diagnostic for Contact-EC, showing that coarse EC levels remain predictable while exact Level-4 specificity collapses.

Table 9: Contact-EC prediction-cardinality and confidence audit at threshold 0.5. Partial and novel temporal proteins are not fully Level-4 scorable, so their exact-hit and micro-F1 fields are reported as N/A.

| Dataset             | $N$ | Avg. L4 preds | Empty rate | Mean top-1 $p$ | Top-1 hit | L4 micro F1 |
|---------------------|-----|---------------|------------|----------------|-----------|-------------|
| Temporal known EC   | 124 | 0.7984        | 36.3%      | 0.6657         | 57.3%     | 0.6032      |
| Temporal partial EC | 309 | 0.4595        | 64.4%      | 0.3862         | N/A       | N/A         |
| Temporal novel EC   | 35  | 0.4286        | 74.3%      | 0.3404         | N/A       | N/A         |
| Price-149           | 136 | 0.6471        | 48.5%      | 0.5187         | 27.9%     | 0.2500      |

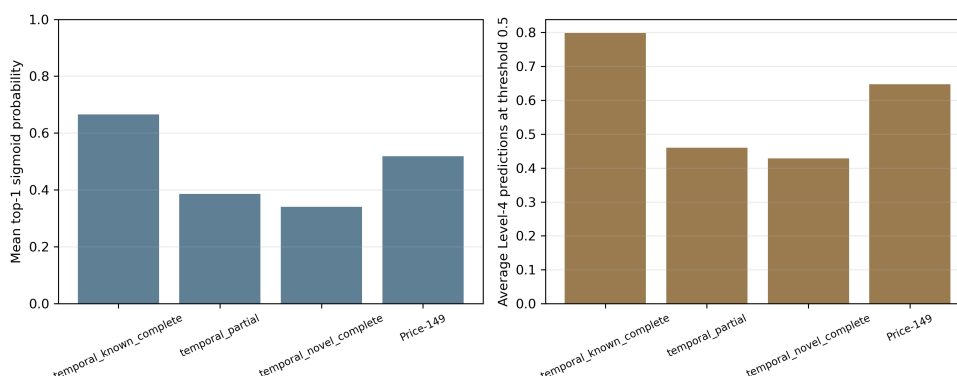


Figure 5: Contact-EC calibration and prediction-cardinality diagnostics. The audit separates empty prediction rate, average predicted-label cardinality, top-1 confidence, and top-1 correctness for temporal known, temporal partial, temporal novel, and Price-149 proteins.

**Late probability fusion baselines.** To test whether Contact-EC’s gain could be reproduced by simple post-hoc combination of independently trained sequence-only and contact-only models, we evaluated probability-level fusion baselines using B1 and B3 predictions: mean posterior averaging, element-wise posterior maximum, and two fixed weighted averages. These baselines require no retraining and therefore provide a conservative diagnostic rather than a full architectural ablation. As shown in Table 10 and Figure 6, learned Contact-EC substantially outperforms all late-fusion variants on the 124 complete temporal proteins (0.6032 vs. 0.4673 for the best late-fusion baseline). The same pattern holds on Price-149, where late fusion remains near the weak ESM-only baseline while Contact-EC reaches 0.2500. Thus the observed sequence–structure gain is not explained by naive posterior averaging or maximum pooling.

Table 10: Late probability-fusion baselines using independently trained B1 (ESM-2) and B3 (contact-only) models. Values are Level-4 micro F1 at threshold 0.5.

| Model                     | Swiss-Prot 2023 known EC ( $N = 124$ ) | Price-149 ( $N = 136$ ) |
|---------------------------|----------------------------------------|-------------------------|
| B1 (ESM-2)                | 0.4216                                 | 0.0324                  |
| B3 (Contact)              | 0.3646                                 | 0.0000                  |
| Late mean B1+B3           | 0.3770                                 | 0.0118                  |
| Late max B1+B3            | 0.4673                                 | 0.0312                  |
| Late weighted 0.7B1+0.3B3 | 0.4322                                 | 0.0231                  |
| Late weighted 0.3B1+0.7B3 | 0.3598                                 | 0.0000                  |
| Contact-EC learned fusion | <b>0.6032</b>                          | <b>0.2500</b>           |

**Trainable fusion architecture controls.** We also trained three same-encoder, flat-head sequence–structure fusion controls on the SP-2018 EC-Bench training corpus and evaluated them on the complete Swiss-Prot 2023-01 temporal subset. The concat control concatenates projected ESM-2 and contact-map features before an MLP, the sum control adds the two projected features, and the gated-MLP control learns a feature-wise interpolation between the two projections. Unlike the late-fusion baselines, these controls are trained end to end over the downstream fusion layers and therefore directly test whether Contact-EC’s gated additive fusion is merely a generic benefit of adding contact-map features. Table 11 shows that Contact-EC remains strongest across three seeds, although concat fusion is close enough that the architecture should be interpreted as a useful controlled fusion choice rather than a large standalone architectural breakthrough.

Table 11: Trainable simple fusion baselines on the complete Swiss-Prot 2023-01 temporal subset ( $N = 124$ ). Values are mean  $\pm$  sample standard deviation across seeds 42/43/44. All trainable controls use the same contact encoder family and flat Level-4 prediction head as Contact-EC.

| Model              | Micro F1                              | Weighted F1                           | Precision           | Recall                                | Rare-EC F1                            |
|--------------------|---------------------------------------|---------------------------------------|---------------------|---------------------------------------|---------------------------------------|
| Concat flat FC     | 0.5922 $\pm$ 0.0150                   | 0.4777 $\pm$ 0.0116                   | 0.8269 $\pm$ 0.0479 | 0.4619 $\pm$ 0.0136                   | 0.4729 $\pm$ 0.0113                   |
| Sum flat FC        | 0.5505 $\pm$ 0.0230                   | 0.4818 $\pm$ 0.0067                   | 0.6638 $\pm$ 0.0433 | 0.4706 $\pm$ 0.0131                   | 0.3970 $\pm$ 0.0112                   |
| Gated MLP flat FC  | 0.5543 $\pm$ 0.0525                   | 0.4532 $\pm$ 0.0413                   | 0.7768 $\pm$ 0.0692 | 0.4314 $\pm$ 0.0458                   | 0.4132 $\pm$ 0.0368                   |
| Contact-EC flat FC | <b>0.6241 <math>\pm</math> 0.0170</b> | <b>0.5278 <math>\pm</math> 0.0222</b> | 0.7997 $\pm$ 0.0226 | <b>0.5120 <math>\pm</math> 0.0200</b> | <b>0.5257 <math>\pm</math> 0.0313</b> |

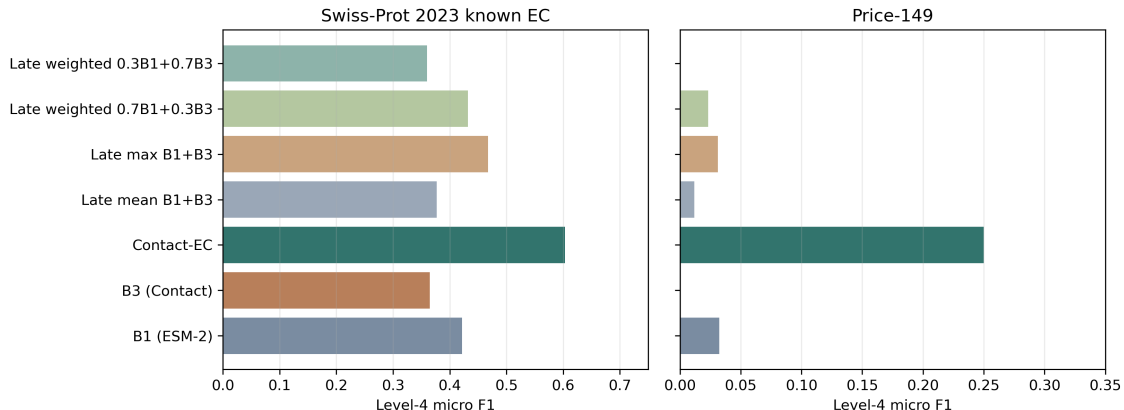


Figure 6: Late probability-fusion baselines. Learned Contact-EC exceeds simple B1/B3 posterior averaging, posterior maximum, and fixed weighted averages on both the temporal known-EC subset and Price-149.

Table 12: Contact map channel perturbation (B3 checkpoint, random-split test,  $N = 22,018$  complete-L4 proteins). Inference-time zeroing of channels 1–2 is a pessimistic lower bound on the training-time 1-channel ablation.

| Input channels                                  | L4 Micro F1   | $\Delta$ vs. 3ch |
|-------------------------------------------------|---------------|------------------|
| 3-channel (all / short / long)                  | <b>0.8684</b> | —                |
| 1-channel <sup>†</sup> (ch0 only, ch1–2 zeroed) | 0.1187        | −74.97 pp        |

## S6 Contact Map Channel Ablation

We encode contact maps as three channels—all contacts, short-range ( $|i-j| < 12$ ), and long-range ( $|i-j| \geq 12$ ). Inference-time zeroing of channels 1–2 drops L4 micro F1 from 0.8684 to 0.1187 (−74.97 pp), confirming the ResNet encoder exploits the channel decomposition to distinguish secondary from tertiary structure.

## S7 Grad-CAM Structural Interpretability

We apply Gradient-weighted Class Activation Mapping (Grad-CAM) [Selvaraju et al., 2017] to the ResNet-50 contact map branch to verify that Contact-EC encodes structurally meaningful contact patterns rather than superficial statistics.

For **Lysozyme C** (EC 3.2.1.17; UniProt C0HLB7), Grad-CAM highlights two discrete regions centred around residues 85–140 and 180–220, corresponding to the catalytic  $\beta$ -sheet strand and  $\alpha$ -helix cluster adjacent to the Glu35–Asp52 active-site dyad [Blake et al., 1967].

For **Phospholipase A1** (EC 3.1.1.32; UniProt P0DPT0), a single hot spot emerges around residues 110–175, co-localising with the serine–histidine–aspartate catalytic triad of the  $\alpha/\beta$  hydrolase fold [Schrage and Cygler, 1993].

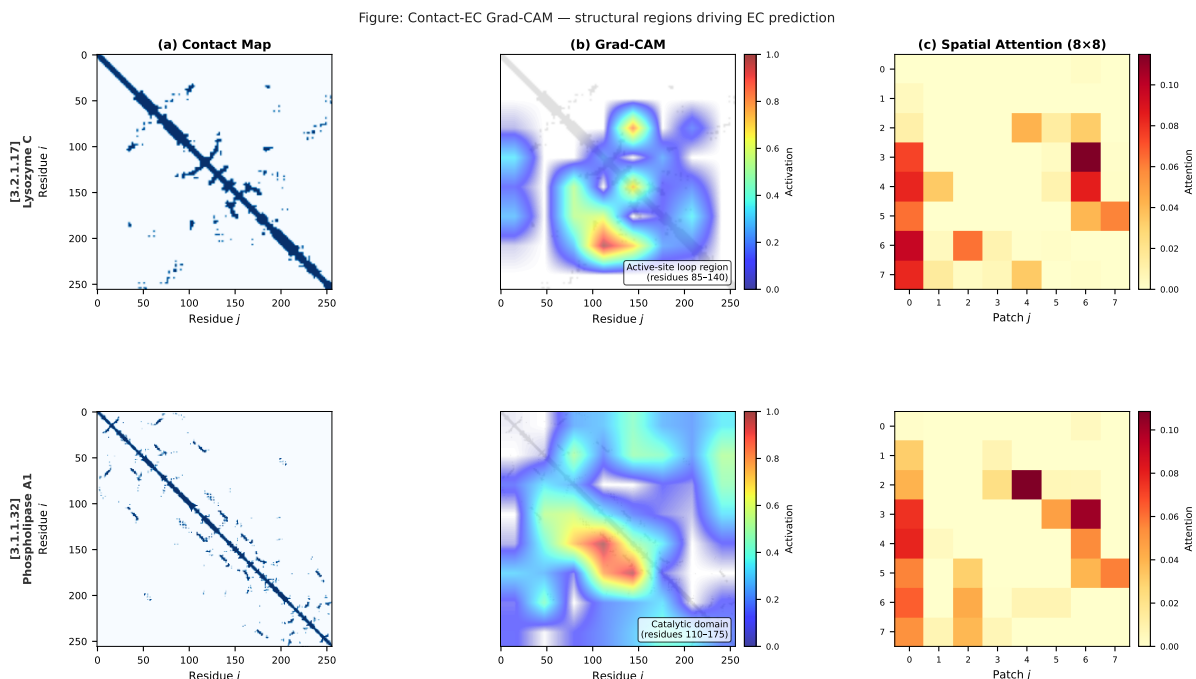


Figure 7: **Grad-CAM structural interpretability for Contact-EC.** Two correctly predicted test-set enzymes. **(a)** AlphaFold-derived contact map. **(b)** Grad-CAM heatmap overlaid on the contact map. **(c)** ResNet-50 spatial attention weights at the  $8 \times 8$  feature-map level. *Top*: Lysozyme C (EC 3.2.1.17, UniProt C0HLB7). *Bottom*: Phospholipase A1 (EC 3.1.1.32, UniProt P0DPT0).

## S8 Extended Discussion

**Comparison to CLEAN-Contact.** CLEAN-Contact uses ESM-2 3B (2,560-dimensional) with contrastive kNN inference and full Swiss-Prot training data. On New-392, Contact-EC (0.3717, SP-2018) trails CLEAN-Contact (0.566) due to backbone scale (3B vs. 650M) and training recency. Contact-EC-ExpA (SP-2026, 0.7680) surpasses CLEAN-Contact but with potential train/test overlap on New-392. On the Swiss-Prot 2023 temporal audit, ExpA is instead evaluated on the  $N = 99$  encoder-evaluable intersection subset, where it improves over the matched SP-2018 Contact-EC baseline by +9.5 pp micro F1 across three seeds. The gap on Price-149 reflects the OOD nature of bacterial RefSeq sequences and exact Level-4 calibration under structure-source shift rather than a simple missing-input problem.

**Paper significance and scope.** The current results demonstrate that structural contact maps provide complementary and synergistic information alongside PLM embeddings, empirically supported on rigorous OOD evaluation. The structural parity finding ( $B3 \approx B1$  on hard OOD validation) is, to our knowledge, a novel empirical observation: prior structure-aware EC methods either use contact maps with much larger sequence models or treat structure as a minor addition.

**Threshold calibration and temporal generalization.** At inference, we apply a fixed sigmoid threshold of 0.5 for Level-4 EC predictions. The sensitivity analysis in Table 6 indicates that this cutoff is reasonably stable for micro F1 but trades precision against recall. The gap in Rare-EC F1 between Contact-EC (0.4219) and Contact-EC-Hier (0.3621) further suggests that the hierarchical head suppresses rare-class outputs through cross-level attention. Future work should explore class-adaptive thresholding and

calibrated decision rules to improve rare-class recall without increasing false positive assignments.

**3B backbone scaling.** Contact-EC-3B (ESM-2 3B backbone, flat FC head, Phase 1 only) achieves micro F1 = 0.6316 on the temporal test, +2.8 pp over the 650M model. The modest gain for a  $4.5\times$  parameter increase suggests the model is data-constrained rather than backbone-constrained at the current training scale.

## S9 Reproducibility and Audit Checks

All reported splits were audited for accession overlap, exact sequence overlap, label coverage, multi-label cardinality, and contact-map availability. The audit scripts generate machine-readable CSV summaries used to construct the final result tables. Random-split, cluster-split, EC-Bench-style, and external benchmark results are labeled separately to avoid mixing evaluation protocols.

Table 13: Final temporal leakage and label-encoder audit for the complete-known Swiss-Prot 2023-01 proteins. Accession and exact-sequence overlap are necessary checks but do not establish homolog- or fold-disjointness. ExpA contains many temporal proteins in the full corpus because it is a later Swiss-Prot snapshot, but the training split excludes them; only 99 temporal proteins are Level-4 evaluable under the ExpA encoder.

| Training corpus              | Train $N$ | Eval. $N$ | Corpus acc. ovl. | Train acc. ovl. | Train seq. ovl. | Encoder L4 | Train L4 |
|------------------------------|-----------|-----------|------------------|-----------------|-----------------|------------|----------|
| SP-2018 EC-Bench train       | 201,865   | 124       | 0                | 0               | 0               | 4,647      | 4,521    |
| SP-2022 train                | 243,335   | 124       | 0                | 0               | 0               | 5,306      | 5,173    |
| ExpA recent Swiss-Prot train | 243,198   | 99        | 115              | 0               | 0               | 5,306      | 5,164    |

Table 14: Matched temporal recency comparison on the Level-4 encoder-evaluable intersection subset ( $N = 99$ ). This avoids comparing ExpA against the broader SP-2018-only  $N = 124$  denominator.

| Model                   | Runs    | Micro F1            | Weighted F1         | Precision           | Recall              |
|-------------------------|---------|---------------------|---------------------|---------------------|---------------------|
| Contact-EC SP-2018      | 3 seeds | $0.6467 \pm 0.0210$ | $0.5692 \pm 0.0214$ | $0.7873 \pm 0.0175$ | $0.5489 \pm 0.0233$ |
| Contact-EC-ExpA SP-2026 | 3 seeds | $0.7417 \pm 0.0182$ | $0.6768 \pm 0.0168$ | $0.8583 \pm 0.0096$ | $0.6532 \pm 0.0233$ |

Table 15: MMseqs2 nearest-neighbour and label-coverage audit for the temporal recency comparison. Searches use complete Level-4 EC-labelled proteins in each training split as the target database. The newer corpora increase nearest-neighbour identity and top-hit exact EC agreement on the matched  $N = 99$  subset, so the ExpA gain should be interpreted as a recent-corpus effect that includes homolog availability and label-frequency changes rather than a pure calendar-date effect.

| Subset               | Training corpus | $N$ | Median top identity | $\geq 0.60$ identity | $\geq 0.90$ identity | All L4 in train | Top-hit same L4 |
|----------------------|-----------------|-----|---------------------|----------------------|----------------------|-----------------|-----------------|
| Complete-known       | SP-2018         | 124 | 0.555               | 46                   | 9                    | 113             | 85              |
| Complete-known       | SP-2022         | 124 | 0.644               | 63                   | 18                   | 115             | 95              |
| Complete-known       | SP-2026 ExpA    | 124 | 0.644               | 63                   | 18                   | 114             | 95              |
| Matched intersection | SP-2018         | 99  | 0.556               | 38                   | 8                    | 99              | 69              |
| Matched intersection | SP-2022         | 99  | 0.619               | 49                   | 14                   | 99              | 78              |
| Matched intersection | SP-2026 ExpA    | 99  | 0.619               | 49                   | 14                   | 99              | 78              |

**Sequence-similarity stratification of temporal cases.** To test whether temporal performance is concentrated in proteins with close training-set neighbours, we joined the final case-wise temporal comparison

( $N = 124$ ) with the available EC-Bench test-vs-train similarity audit. This similarity file covers 101 of the 124 fully evaluable temporal proteins; the remaining 23 proteins are therefore excluded from the stratified analysis. Table 16 and Figure 8 show that Contact-EC performance is substantially lower when no recorded sequence similarity to the training set is available, and increases in the 0.60–0.90 identity regime. This supports the manuscript’s conservative interpretation: temporal EC performance is sensitive to train-test relatedness, and sequence-disjoint validation does not by itself establish fold-disjoint structural generalization.

Table 16: Temporal performance stratified by maximum recorded train-set sequence identity for the 101 temporal proteins with available EC-Bench similarity metadata. This is a sequence-similarity sensitivity audit, not a structural fold-disjoint evaluation.

| Max train identity | $N$ | Contact-EC F1 | HIT-EC F1 | Contact hit | HIT-EC hit |
|--------------------|-----|---------------|-----------|-------------|------------|
| 0.00               | 32  | 0.3673        | 0.8125    | 0.2812      | 0.8125     |
| (0.30, 0.60]       | 36  | 0.5902        | 0.8108    | 0.5000      | 0.8333     |
| (0.60, 0.90]       | 27  | 0.8889        | 0.9643    | 0.8889      | 1.0000     |
| >0.90              | 6   | 0.8000        | 0.8333    | 0.6667      | 0.8333     |

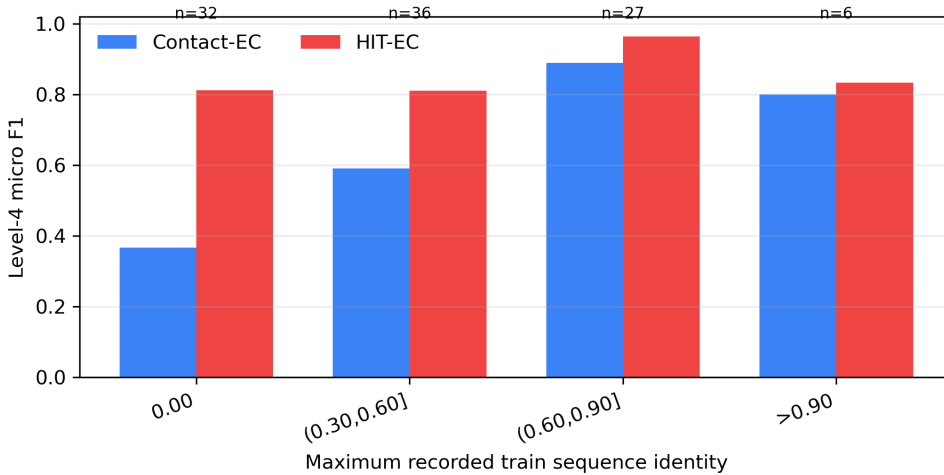


Figure 8: Level-4 micro F1 by maximum recorded train-set sequence identity on the 101 temporal proteins with available EC-Bench similarity metadata. The result motivates the explicit Foldseek-disjoint benchmark reported below.

**Contact-pair nearest-neighbor audit.** Because sequence-disjoint evaluation does not guarantee structural fold-disjointness, we also performed a model-proximal relatedness audit using the existing contact-pair ESM representation. For each protein, we summarized the contact-pair tensor by concatenating the mean and standard deviation across the sampled residue pairs and computed cosine similarity to all processed SP-2018 training proteins with available contact-pair embeddings. This analysis is not a Foldseek, TM-align, CATH, or SCOP fold-disjoint benchmark. Instead, it asks whether temporal test proteins have close neighbours in the same representation space used by the contact branch. Table 17 and Figure 9 show that all 113 evaluable temporal known-EC proteins have a nearest training neighbour with contact-pair cosine similarity above 0.90. The nearest neighbour shares the same Level-4 EC label in 59.3% of cases. We therefore avoid interpreting the temporal results as evidence of fold-disjoint structural

generalization; the Foldseek-disjoint benchmark below addresses this question directly by reclustering the training corpus and retraining the models on the resulting split.

Table 17: Contact-pair nearest-neighbor audit for temporal known-EC proteins. Cosine similarity is computed on mean–standard deviation summaries of existing contact-pair ESM tensors. This is a representation-neighbour audit, not a structural fold-disjoint benchmark.

| Quantity                                                | Value           |
|---------------------------------------------------------|-----------------|
| Temporal known-EC proteins with contact-pair embeddings | 113/124         |
| SP-2018 training proteins with contact-pair embeddings  | 212,863/224,294 |
| Median nearest-neighbour cosine                         | 0.9852          |
| Mean nearest-neighbour cosine                           | 0.9845          |
| Temporal proteins with nearest cosine $>0.90$           | 113/113         |
| Nearest neighbour same Level-1 EC                       | 87.6%           |
| Nearest neighbour same Level-2 EC                       | 82.3%           |
| Nearest neighbour same Level-3 EC                       | 82.3%           |
| Nearest neighbour same Level-4 EC                       | 59.3%           |

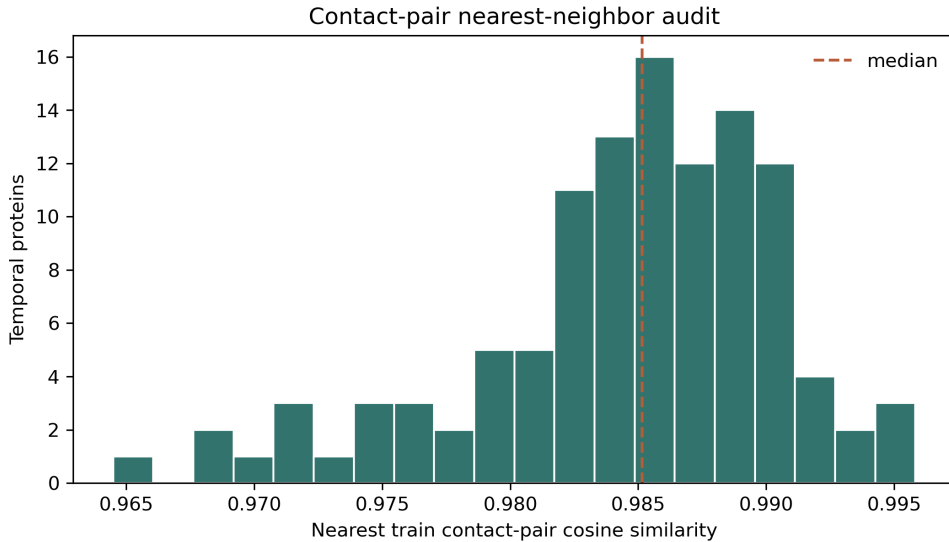


Figure 9: Distribution of nearest SP-2018 training-set cosine similarity for the 113 temporal known-EC proteins with available contact-pair embeddings. The high nearest-neighbour similarities indicate that the current temporal evaluation should not be treated as fold-disjoint structural generalization.

**Raw contact-map topology nearest-neighbor audit.** To make the structural-relatedness audit less dependent on the learned contact-pair ESM representation, we repeated the nearest-neighbor analysis using hand-crafted fingerprints derived directly from the  $256 \times 256$  contact maps. Each fingerprint concatenates a  $32 \times 32$  coarse contact grid, sequence-separation contact densities, and node-degree distribution statistics ( $d = 1041$ ). This is still not a Foldseek/TM-align or CATH/SCOP fold-disjoint benchmark, but it uses the raw structural input rather than a model-internal representation. Table 18 and Figure 10 show that 114/115 temporal proteins with available contact maps have a nearest SP-2018 training protein with topology cosine similarity  $>0.90$ , and 63/115 exceed 0.98. The nearest topology neighbour shares the same Level-4 EC label in 40.9% of cases, rising to 57.1% among proteins with cosine similarity  $>0.98$ . This independent audit reinforces the conservative interpretation: the temporal split

measures annotation-time shift and label-vocabulary shift, but it should not be presented as fold-disjoint structural generalization.

Table 18: Raw contact-map topology nearest-neighbor audit for temporal known-EC proteins. Similarity is computed on hand-crafted fingerprints from the  $256 \times 256$  contact maps, independent of the learned contact-pair ESM representation.

| Quantity                                      | Value           |
|-----------------------------------------------|-----------------|
| Temporal known-EC proteins with contact maps  | 115/124         |
| SP-2018 training proteins with contact maps   | 212,863/224,294 |
| Fingerprint dimension                         | 1,041           |
| Median nearest-neighbour cosine               | 0.9822          |
| Mean nearest-neighbour cosine                 | 0.9733          |
| Temporal proteins with nearest cosine $>0.90$ | 114/115         |
| Temporal proteins with nearest cosine $>0.95$ | 93/115          |
| Temporal proteins with nearest cosine $>0.98$ | 63/115          |
| Nearest neighbour same Level-1 EC             | 75.7%           |
| Nearest neighbour same Level-2 EC             | 69.0%           |
| Nearest neighbour same Level-3 EC             | 69.0%           |
| Nearest neighbour same Level-4 EC             | 40.9%           |

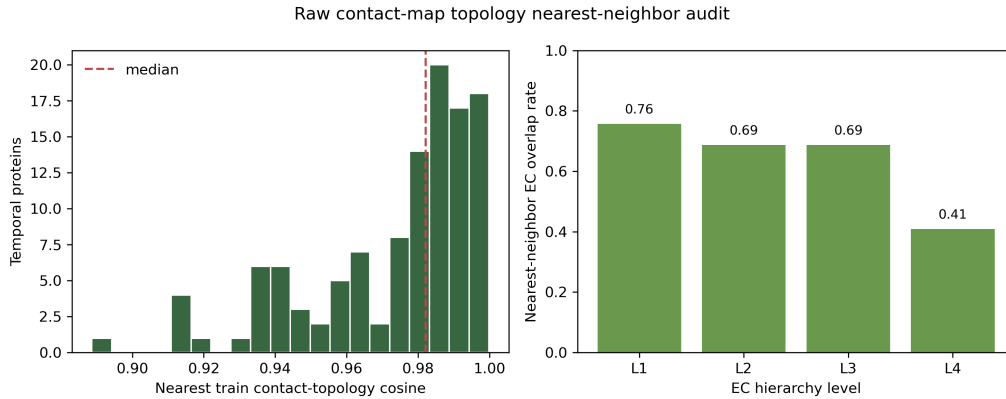


Figure 10: Raw contact-map topology nearest-neighbor audit. Left: nearest SP-2018 training-set cosine similarity for temporal known-EC proteins with available contact maps. Right: EC hierarchy overlap with the nearest topology neighbour.

**Foldseek/TM-score-disjoint benchmark.** We constructed an explicit fold-level split by clustering all SP-2018 proteins with available AlphaFold structures using Foldseek [van Kempen et al., 2024]. The clustering used TM-score threshold 0.50, coverage 0.80, minimum sequence identity 0.0, alignment type 1, cluster mode 1, and sensitivity 7.5. Of 224,294 metadata proteins, 212,863 had available structures and were assigned to 3,813 clusters. The cluster-count-balanced split contains 170,573 train proteins, 25,565 validation proteins, and 16,725 test proteins, with 3,051/381/381 structure clusters and zero protein-ID overlap across partitions.

The Foldseek split also changes label coverage. Among Foldseek-test positive Level-4 labels, 31.5% are absent from the corresponding Foldseek training split, and 31.6% of test proteins contain at least one unseen Level-4 label. Therefore, fold-disjoint evaluation mixes structural shift with closed-set vocabulary and frequency shift; the latter must be reported rather than attributed solely to fold-level generalization.



Table 19: Foldseek/TM-score-disjoint split audit. Missing-structure proteins are listed in the released audit files and excluded from this structure-disjoint training/evaluation split.

| Quantity                           | Value                |
|------------------------------------|----------------------|
| Metadata proteins                  | 224,294              |
| Proteins with available structures | 212,863              |
| Missing structures                 | 11,431               |
| Foldseek clusters                  | 3,813                |
| Cluster size median / mean / max   | 2.0 / 55.83 / 18,692 |
| Train proteins / clusters          | 170,573 / 3,051      |
| Validation proteins / clusters     | 25,565 / 381         |
| Test proteins / clusters           | 16,725 / 381         |
| Train-validation ID overlap        | 0                    |
| Train-test ID overlap              | 0                    |
| Validation-test ID overlap         | 0                    |

Table 20: Foldseek/TM-score-disjoint Level-4 performance for the representative base checkpoints. The fixed threshold 0.5 is the default sigmoid cutoff. Validation-selected thresholds are selected on the Foldseek validation split and applied once to the held-out Foldseek test split.

| Model               | F1@0.5        | Val thr. | Test F1@val thr. | Precision     | Recall        |
|---------------------|---------------|----------|------------------|---------------|---------------|
| B1 (ESM-2, flat FC) | 0.0452        | 0.130    | 0.0613           | 0.1407        | 0.0392        |
| B3 (contact only)   | 0.0436        | 0.070    | 0.0652           | 0.0619        | 0.0690        |
| Contact-EC (fusion) | <b>0.0864</b> | 0.080    | <b>0.1747</b>    | <b>0.2406</b> | <b>0.1371</b> |

Table 21: Foldseek/TM-score sweep split composition. The 0.50 split was generated with cluster-count assignment, whereas the 0.40 and 0.60 splits were generated with protein-balanced assignment. Thus the sweep is a robustness/composition audit rather than a strictly monotonic threshold-only curve.

| TM-score | Assignment       | Clusters | Max cluster | Test proteins | Unseen pos. | Any-unseen proteins |
|----------|------------------|----------|-------------|---------------|-------------|---------------------|
| 0.40     | protein-balanced | 10,893   | 1,026       | 21,617        | 0.159       | 0.170               |
| 0.50     | cluster-count    | 3,813    | 18,692      | 16,725        | 0.315       | 0.316               |
| 0.60     | protein-balanced | 11,869   | 1,026       | 21,555        | 0.103       | 0.110               |

Table 22: Single-seed neural model performance across Foldseek/TM-score sweep splits at the fixed Level-4 threshold of 0.5. Fusion is consistently strongest, but absolute scores must be interpreted together with the split-composition audit in Table 21.

| TM-score split | B1 ESM-2 | B3 contact | Contact-EC fusion |
|----------------|----------|------------|-------------------|
| 0.40           | 0.0683   | 0.2622     | <b>0.2981</b>     |
| 0.50           | 0.0462   | 0.0337     | <b>0.0998</b>     |
| 0.60           | 0.0538   | 0.3985     | <b>0.4403</b>     |

Table 23: MMseqs2 top-hit homology-transfer baseline on Foldseek/TM-score split variants. The baseline transfers all Level-4 EC labels from the single best MMseqs2 hit in the corresponding training partition. Higher TM-score thresholds produce stricter clustering in the newly generated threshold sweep.

| Split         | Coverage | L1 hit | L2 hit | L3 hit | L4 hit | L4 micro F1 |
|---------------|----------|--------|--------|--------|--------|-------------|
| TM-score 0.40 | 0.902    | 0.865  | 0.798  | 0.766  | 0.660  | 0.694       |
| TM-score 0.50 | 0.753    | 0.706  | 0.642  | 0.614  | 0.473  | 0.539       |
| TM-score 0.60 | 0.962    | 0.938  | 0.933  | 0.933  | 0.802  | 0.818       |

Table 24: MMseqs2 top-hit homology-transfer baseline on temporal and external test sets using the SP-2018 training corpus as the search database. Price-149 is reported both on the encoded subset used by the neural models and on the raw 149-protein benchmark.

| Test set                               | Coverage | L1 hit | L2 hit | L3 hit | L4 hit | L4 micro F1 |
|----------------------------------------|----------|--------|--------|--------|--------|-------------|
| Swiss-Prot 2023 known EC ( $N = 124$ ) | 0.871    | 0.831  | 0.831  | 0.831  | 0.637  | 0.585       |
| Price-149 encoded ( $N = 136$ )        | 0.963    | 0.941  | 0.875  | 0.809  | 0.338  | 0.345       |
| Price-149 raw ( $N = 149$ )            | 0.960    | 0.940  | 0.872  | 0.805  | 0.309  | 0.312       |

Table 25: Three-seed temporal known-EC Level-4 performance on the complete Swiss-Prot 2023-01 subset ( $N = 124$ ). Values are mean  $\pm$  sample standard deviation across seeds 42/43/44.

| Model               | Micro F1                              | Weighted F1                           | Precision                             | Recall                                | Rare-EC F1                            |
|---------------------|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|
| B1 (ESM-2, flat FC) | 0.4508 $\pm$ 0.0203                   | 0.3387 $\pm$ 0.0194                   | <b>0.9035 <math>\pm</math> 0.0445</b> | 0.3006 $\pm$ 0.0173                   | 0.3292 $\pm$ 0.0272                   |
| B3 (contact only)   | 0.4244 $\pm$ 0.0207                   | 0.3325 $\pm$ 0.0237                   | 0.8300 $\pm$ 0.0220                   | 0.2854 $\pm$ 0.0200                   | 0.3354 $\pm$ 0.0254                   |
| Contact-EC flat FC  | <b>0.6241 <math>\pm</math> 0.0170</b> | <b>0.5278 <math>\pm</math> 0.0222</b> | 0.7997 $\pm$ 0.0226                   | <b>0.5120 <math>\pm</math> 0.0200</b> | <b>0.5257 <math>\pm</math> 0.0313</b> |

Table 26: MMseqs2 top-hit performance by percent-identity bin on the primary TM-score 0.50 Foldseek split. This analysis shows that Foldseek-disjoint evaluation still contains substantial sequence-homology signal for many test proteins.

| Top-hit identity bin | Proteins | Fraction | L4 hit | L4 micro F1 |
|----------------------|----------|----------|--------|-------------|
| No hit               | 4,138    | 0.247    | 0.000  | 0.000       |
| 0–20%                | 480      | 0.029    | 0.250  | 0.250       |
| 20–30%               | 3,748    | 0.224    | 0.466  | 0.466       |
| 30–50%               | 6,849    | 0.410    | 0.677  | 0.677       |
| 50–70%               | 857      | 0.051    | 0.903  | 0.903       |
| $\geq 70\%$          | 653      | 0.039    | 0.960  | 0.960       |

Table 27: Three-seed Foldseek/TM-score-disjoint Level-4 performance. Values are mean  $\pm$  sample standard deviation over the base checkpoint and two additional training seeds. Validation thresholds are selected on the Foldseek validation split and applied once to the held-out Foldseek test split.

| Model               | F1@0.5                                | Test F1@val thr.                      | Post-hoc best F1                      |
|---------------------|---------------------------------------|---------------------------------------|---------------------------------------|
| B1 (ESM-2, flat FC) | 0.0433 $\pm$ 0.0042                   | 0.0607 $\pm$ 0.0009                   | 0.0666 $\pm$ 0.0042                   |
| B3 (contact only)   | 0.0400 $\pm$ 0.0054                   | 0.0654 $\pm$ 0.0016                   | 0.0700 $\pm$ 0.0036                   |
| Contact-EC (fusion) | <b>0.0872 <math>\pm</math> 0.0122</b> | <b>0.1685 <math>\pm</math> 0.0334</b> | <b>0.1769 <math>\pm</math> 0.0378</b> |

Table 28: Three-seed Foldseek-disjoint top- $k$  diagnostics. Hit rate counts whether any true Level-4 EC label appears among the top- $k$  predicted labels; values are mean  $\pm$  sample standard deviation.

| Model               | Top-1 hit                             | Top-3 hit                             | Top-5 hit                             | Top-10 hit                            |
|---------------------|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|
| B1 (ESM-2, flat FC) | 0.0503 $\pm$ 0.0013                   | 0.0977 $\pm$ 0.0029                   | 0.1203 $\pm$ 0.0015                   | 0.1455 $\pm$ 0.0043                   |
| B3 (contact only)   | 0.0583 $\pm$ 0.0089                   | 0.1105 $\pm$ 0.0051                   | 0.1317 $\pm$ 0.0102                   | 0.1566 $\pm$ 0.0093                   |
| Contact-EC (fusion) | <b>0.1383 <math>\pm</math> 0.0381</b> | <b>0.2445 <math>\pm</math> 0.0489</b> | <b>0.2778 <math>\pm</math> 0.0500</b> | <b>0.3257 <math>\pm</math> 0.0434</b> |

Table 29: Foldseek-disjoint failure-mode audit across three seeds. A rescue means that fusion predicts at least one true Level-4 EC label for a protein for which the comparison baseline has zero sample-level hit rate across seeds.

| Quantity                                       | Fraction of Foldseek-test proteins |
|------------------------------------------------|------------------------------------|
| Fusion rescue vs. B1                           | 0.173                              |
| Fusion rescue vs. B3                           | 0.125                              |
| Fusion-only rescue vs. both B1 and B3          | 0.110                              |
| Missed by all three models                     | 0.753                              |
| All-model failures with any unseen true label  | 0.419                              |
| Fusion-only rescues with any unseen true label | 0.005                              |

Table 30: Representative Foldseek-disjoint fusion-only rescue cases. These examples have seen true Level-4 labels in the Foldseek training partition, zero hit rate for both single-modality baselines across seeds, and nonzero fusion hit rate.

| Accession | EC family       | True EC   | Frequency bin | Train count | Fusion hit | Fusion top-1 |
|-----------|-----------------|-----------|---------------|-------------|------------|--------------|
| P43418    | Hydrolases      | 3.1.21.4  | 26–100        | 95          | 1.000      | 1.000        |
| A5F3I6    | Oxidoreductases | 1.8.4.8   | 6–25          | 8           | 1.000      | 1.000        |
| P51040    | Transferases    | 2.3.3.16  | 6–25          | 9           | 1.000      | 1.000        |
| Q9CB92    | Lyases          | 4.2.99.18 | 26–100        | 54          | 0.667      | 0.667        |
| Q9ZFH0    | Isomerases      | 5.1.3.37  | 6–25          | 7           | 0.667      | 1.000        |

Table 31: Foldseek-disjoint open-vocabulary confidence audit. The target is whether a test protein contains at least one true Level-4 label absent from the Foldseek training split. Scores are unsupervised confidence diagnostics averaged over three seeds; higher AUROC/AP indicates better detection of unseen-label cases.

| Model               | Best confidence score     | AUROC        | AP           |
|---------------------|---------------------------|--------------|--------------|
| B1 (ESM-2, flat FC) | Low top-5 probability sum | 0.676        | 0.436        |
| B3 (contact only)   | Low top-1 probability     | 0.574        | 0.365        |
| Contact-EC (fusion) | Low top-5 probability sum | <b>0.738</b> | <b>0.590</b> |

Table 32: Foldseek-disjoint abstention and parent-level fallback audit for three-seed averaged Contact-EC fusion probabilities. Proteins are abstained in ascending order of top-5 Level-4 probability sum. Retained Level-4 micro F1 is computed only on accepted predictions; abstained Level-3 top-1 hit evaluates a conservative parent-level fallback for deferred proteins.

| Abstained | Coverage | Retained L4 micro F1 | Abstained unseen prevalence | Abstained L3 top-1 hit |
|-----------|----------|----------------------|-----------------------------|------------------------|
| 0%        | 1.00     | 0.208                | –                           | –                      |
| 10%       | 0.90     | 0.222                | 0.712                       | 0.059                  |
| 20%       | 0.80     | 0.239                | 0.592                       | 0.112                  |
| 30%       | 0.70     | 0.257                | 0.533                       | 0.157                  |
| 40%       | 0.60     | 0.279                | 0.496                       | 0.185                  |
| 50%       | 0.50     | 0.305                | 0.465                       | 0.204                  |

Table 33: Foldseek-disjoint label-frequency decomposition. Values are Level-4 micro F1 at model-specific validation-selected thresholds, stratified by the number of Foldseek-train proteins carrying each true Level-4 label.

| Model               | unseen | 1–5    | 6–25          | 26–100        | >100          |
|---------------------|--------|--------|---------------|---------------|---------------|
| B1 (ESM-2, flat FC) | 0.0000 | 0.0000 | 0.0000        | 0.0847        | 0.1242        |
| B3 (contact only)   | 0.0000 | 0.0055 | 0.0448        | 0.1081        | 0.1825        |
| Contact-EC (fusion) | 0.0000 | 0.0038 | <b>0.2484</b> | <b>0.3278</b> | <b>0.2167</b> |

Table 34: Foldseek-disjoint sample-level hit rate by true-label training frequency. Values are averaged over three seeds using validation-selected thresholds. The fusion gain is concentrated in seen labels, especially 6–100 training examples, whereas unseen labels remain essentially unsolved by closed-set models.

| True-label bin | $N$   | Any unseen | B1 hit | B3 hit | Fusion hit |
|----------------|-------|------------|--------|--------|------------|
| Unseen         | 5,287 | 1.000      | 0.000  | 0.000  | 0.002      |
| 1–5            | 4,025 | 0.000      | 0.000  | 0.020  | 0.031      |
| 6–25           | 2,415 | 0.000      | 0.000  | 0.047  | 0.187      |
| 26–100         | 3,243 | 0.000      | 0.001  | 0.157  | 0.193      |
| >100           | 1,755 | 0.000      | 0.376  | 0.275  | 0.537      |

Table 35: Foldseek-disjoint EC Level-1 family performance at model-specific validation-selected thresholds.

| Family          | B1 F1         | B3 F1  | Fusion F1     |
|-----------------|---------------|--------|---------------|
| Oxidoreductases | 0.1157        | 0.0507 | <b>0.2612</b> |
| Transferases    | 0.0635        | 0.0606 | <b>0.1497</b> |
| Hydrolases      | 0.0777        | 0.0967 | <b>0.2576</b> |
| Lyases          | <b>0.1288</b> | 0.1135 | 0.0468        |
| Isomerases      | <b>0.2468</b> | 0.2440 | 0.2318        |
| Ligases         | 0.0011        | 0.0005 | <b>0.0036</b> |

Table 36: Interpretation of each evaluation protocol. The same numerical metric can answer different scientific questions depending on split construction and label coverage.

| Protocol                           | Primary question                                                                 | Main caveat                                                                                                                                                                                                                                      |
|------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Random Swiss-Prot split            | In-distribution fitting and calibration                                          | Contains exact sequence duplicates across partitions; not evidence of temporal or sequence-disjoint generalization.                                                                                                                              |
| MMSeqs2 cluster split              | Generalization to sequence-disjoint proteins                                     | Valid only when the evaluated checkpoint is trained on the corresponding cluster-train split.                                                                                                                                                    |
| EC-Bench val_hard                  | Controlled hard validation at $\leq 30\%$ sequence identity                      | Still a validation protocol; temporal deployment claims require held-out future annotations.                                                                                                                                                     |
| Swiss-Prot 2023-01 known-EC subset | Closed-set recognition of temporally new proteins whose labels are in vocabulary | Only 124/468 proteins are fully evaluable at Level 4; contact-pair and raw contact-map audits show high nearest-neighbour structural relatedness.                                                                                                |
| Swiss-Prot 2023-01 full set        | Sensitivity to temporal label shift and incomplete annotations                   | Operational coverage audit only; partial labels are not fully Level-4 scorable, and novel labels are outside the closed-set vocabulary.                                                                                                          |
| New-392 / Price-149                | External stress tests from CLEAN-Contact                                         | New-392 overlaps with current Swiss-Prot; Price-149 is an external bacterial RefSeq stress test with strong Level-4 specificity loss despite available inputs.                                                                                   |
| Foldseek/TM-score-disjoint split   | Closed-set EC prediction under explicit structure-cluster shift                  | Absolute Level-4 performance is low and label coverage varies across generated TM-score sweep splits; the primary 0.50 split has 31.5% unseen positive labels, so results must be interpreted jointly with label coverage and split composition. |

These protocol distinctions are used throughout the manuscript to avoid treating all F1 values as interchangeable. In particular, the Bioinformatics-relevant claim is not that Contact-EC is the strongest available EC predictor in absolute terms. Rather, Contact-EC provides a controlled sequence–structure framework for separating representation effects from temporal cutoff, label-vocabulary shift, sequence overlap, and structure availability.

**Canonical result selection.** When multiple evaluation logs exist for the same checkpoint, we use the full multi-label evaluation on the full evaluable temporal set ( $N = 124$ ) as canonical. Earlier  $N = 101$  evaluations are retained only as historical partial runs and are not used in the main conclusions. For Contact-EC-3B, the canonical temporal result is micro F1 = 0.6316 from the full  $N = 124$  multi-label evaluation; auxiliary argmax-only and partial-set evaluations are excluded from the main result table. For B3 contact-only hard validation, we use the completed resumed checkpoint evaluation (micro F1 = 0.7650, weighted F1 = 0.6788, macro F1 = 0.1335); an earlier interrupted training log reached a higher validation micro F1 before termination but is not used as the paper source because the completed run provides the auditable checkpoint/result pair.

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